

EEE3027 Integrated Circuit Design

Semester 1 Report

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CONTENTS

1	Introduction	3
2	Transistor Design Labs (Labs 1 & 2)	3
2.1	NMOS Sweep and Explanation	3
2.2	PMOS Sweep and Explanation	6
2.3	Task 3, Gain Factor Consolidation and Comparison	7
2.4	Lab 2	16
3	FPGA Design Labs (Labs 4, 5 & 6)	23
4	Asynchronous Circuit Modelling with Workcraft (Labs 7 & 8)	23
5	References	23
6	Appendix	23

Abstract

This is a placeholder for the abstract of the report.

2.1 NMOS Sweep and Explanation

It is important to understand the characteristics and behaviour of NMOS transistors as they are fundamental components in integrated circuit design. In this section, we will discuss the results of the NMOS sweep and provide an explanation of the observed characteristics, as well as the underlying theory behind transistor operation.

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. It is composed of semiconductor material, usually with at least three terminals for connection to an external circuit. A voltage or current applied to one pair of the transistor's terminals controls the current through another pair of terminals.

Transistor Structure:

- **Source (S):** The terminal through which carriers enter the channel.
- **Drain (D):** The terminal through which carriers leave the channel.
- **Gate (G):** The terminal that modulates the conductivity of the channel.
- **Body (B):** The substrate on which the transistor is built, often connected to the source.
- **Channel:** The region between the source and drain where current flows when the transistor is on.
- **Oxide Layer:** An insulating layer between the gate and the channel, typically made of silicon dioxide (SiO_2).



Figure 1: Basic structure of an NMOS and PMOS transistor.

MOSFET Types

- **N-channel (NMOS):** electrons are charge carriers.
- **P-channel (PMOS):** holes are charge carriers.

Operating Regions of a MOSFET

See the table below for the operating regions of an N-channel MOSFET:

Region	Condition (N-channel)	Description
Cutoff	$V_{GS} < V_{th}$	OFF (no current)
Triode (Linear)	$V_{GS} > V_{th}, V_{DS} < V_{GS} - V_{th}$	Acts as variable resistor
Saturation (Active)	$V_{GS} > V_{th}, V_{DS} \geq V_{GS} - V_{th}$	Current source (amplifier region)

Code and Explanation

```
1 * Task 1 Extended: Sweep VDS and VGS (multiple curves)
2 * nmos_DC_sweep.cir is a netlist.
3
4 * Sets the VGS value to 1.2V (can be changed to sweep different values)
5 .param VGSVAL=1.2
6
7 * Tells us the voltages that we will use and the connections (so V_GS
   is a DC source and it has a gate connection (0V DC), same with the
   Drain)
8 VGS G 0 DC {VGSVAL}
9 VDS D 0 DC 0
10 * Tells us the transistor dimensions (gate width (W), channel length (L
    )) along with source and body voltage.
11 M1 D G 0 0 NLEVEL1 W=10u L=1u
12
13 .model NLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
14 * VTO is V_th, KP = transconductance parameter, LAMBDA is the channel-
    length modulation factor.
15
16 .dc VDS 0 1.2 0.01 VGS 0.4 1.2 0.2
17 * plots the drain current (I(M1)s) against VDS for different VGS values
    from 0.4V to 1.2V in steps of 0.2V
18 .plot DC I(M1)s
19 .end
20
21 * Comments start with asterisk
22 * First line is title (can be anything)
23 * Last line must be .END (or .end as SPICE is case insensitive)
24 * Node names: numbers or alphanumeric
25 * Node 0 is always ground
26 * Component values: scientific notation (1e-12) or units (1p, 1n, 1u, 1
    m, 1k, 1meg)
```

Listing 1: NMOS Sweep Simulation Code

The netlist defines an NMOS transistor (M1) with width $10\mu m$ and length $1\mu m$. The `.dc` command performs a nested sweep: the primary sweep varies V_{DS} from $0V$ to $1.2V$ in $10mV$ increments, while the secondary sweep steps V_{GS} from $0.4V$ (threshold) to $1.2V$ in $0.2V$ increments.

This generates five $I - V$ characteristic curves, each corresponding to a different gate overdrive voltage. The Level 1 model parameters define the threshold voltage (VTO), transconductance parameter (KP), and channel-length modulation coefficient (LAMBDA).

Results and Discussion

The resulting plot (**Figure 2**) below shows the drain current (I_D) compared to the drain-source voltage (V_{DS}). 5 distinct curves are visible, each one representing a different gate-source voltage (V_{GS}) from $0.4V$ to $1.2V$.

The simulation produces five curves corresponding to $V_{GS} = 0.4V, 0.6V, 0.8V, 1.0V, \text{ and } 1.2V$.

Each curve exhibits two distinct regions:

Linear Region: For small V_{DS} (where $V_{DS} < V_{GS} - V_{th}$), the drain current increases approximately linearly with V_{DS} . The transistor behaves as a voltage-controlled resistor.

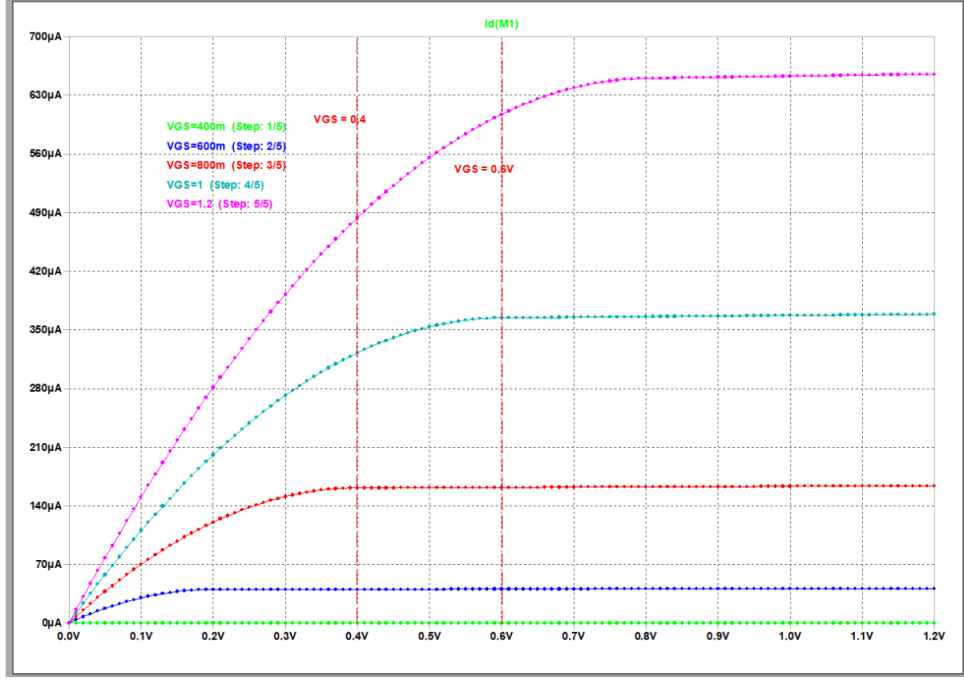


Figure 2: NMOS I-V Characteristics for various V_{GS} values.

Saturation Region: When $V_{DS} \geq V_{GS} - V_{th}$, the channel pinches off and I_D becomes nearly independent of V_{DS} . The slight upward slope in saturation is due to channel-length modulation ($\lambda = 0.02V^{-1}$). Verification Calculation ($V_{GS} = 1.2V$):

Given: $K_P = 200 \mu A/V^2$, $W = 10 \mu m$, $L = 1 \mu m$, $V_{th} = 0.4 V$, and $\lambda = 0.02 V^{-1}$.

$$\beta = K_P \frac{W}{L} \quad (1)$$

$$\beta = 200 \mu \times \frac{10}{1} = 2000 \mu A/V^2$$

At saturation with $V_{DS} = 1.2 V$, the drain current is

$$I_D = \frac{\beta}{2} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (2)$$

Substituting the values,

$$I_D = \frac{2000}{2} (1.2 - 0.4)^2 (1 + 0.02 \times 1.2)$$

Evaluating,

$$I_D = 1000 \times 0.64 \times 1.024 = 655.4 \mu A$$

This closely matches the simulated value, confirming the accuracy of the model.

The effect of channel-length modulation is observed through the parameter λ . If $\lambda = 0$, the saturation characteristics would be perfectly flat. A non-zero value of $\lambda = 0.02 \text{ V}^{-1}$ introduces a finite output resistance given by

$$r_o = \frac{1}{\lambda I_D}. \quad (3)$$

For a drain current of $I_D = 640 \mu\text{A}$,

$$r_o = \frac{1}{0.02 \times 640 \times 10^{-6}} \approx 78 \text{ k}\Omega. \quad (4)$$

This effect is significant in analogue circuit design, as a higher output resistance directly improves amplifier gain.

2.2 PMOS Sweep and Explanation

Procedure and Theory For PMOS, all polarities are reversed relative to NMOS. The source connects to the higher potential, and conduction occurs when $V_{GS} < V_{th}$ (where $V_{th} < 0$).

Region	Condition (P-channel)	Description
Cutoff	$V_{GS} > V_{th}$	OFF (no current)
Triode (Linear)	$V_{GS} < V_{th}, V_{DS} > V_{GS} - V_{th}$	Acts as variable resistor
Saturation (Active)	$V_{GS} < V_{th}, V_{DS} \leq V_{GS} - V_{th}$	Current source (amplifier region)

Code and Explanation

```

1 * Task 2: PMOS I-V DC Sweep with Multiple VGS Curves
2 .param VGSVAL=-1.2
3
4 VGS G 0 DC {VGSVAL}          * Negative gate voltage for PMOS
5 VDS D 0 DC 0
6
7 M1 D G 0 0 PLEVEL1 W=10u L=1u * PMOS transistor
8
9 * Level 1 PMOS model (note negative VTO)
10 .model PLEVEL1 PMOS LEVEL=1 VTO=-0.4 KP=200u LAMBDA=0.02
11
12 * Sweep VDS 0 to -1.2V, VGS from -0.4V to -1.2V
13 .dc VDS 0 -1.2 -0.01 VGS -0.4 -1.2 -0.2
14
15 .plot DC I(M1)
16 .end

```

Listing 2: PMOS Sweep Simulation Code

The netlist mirrors the NMOS structure but with reversed polarities. The *.dc* command sweeps V_{DS} from 0V to -1.2V while stepping V_{GS} from -0.4V (threshold) to -1.2V in -0.2V increments. The PMOS model uses $V_{TO} = -0.4\text{V}$ to reflect the negative threshold voltage.

Results and Discussion

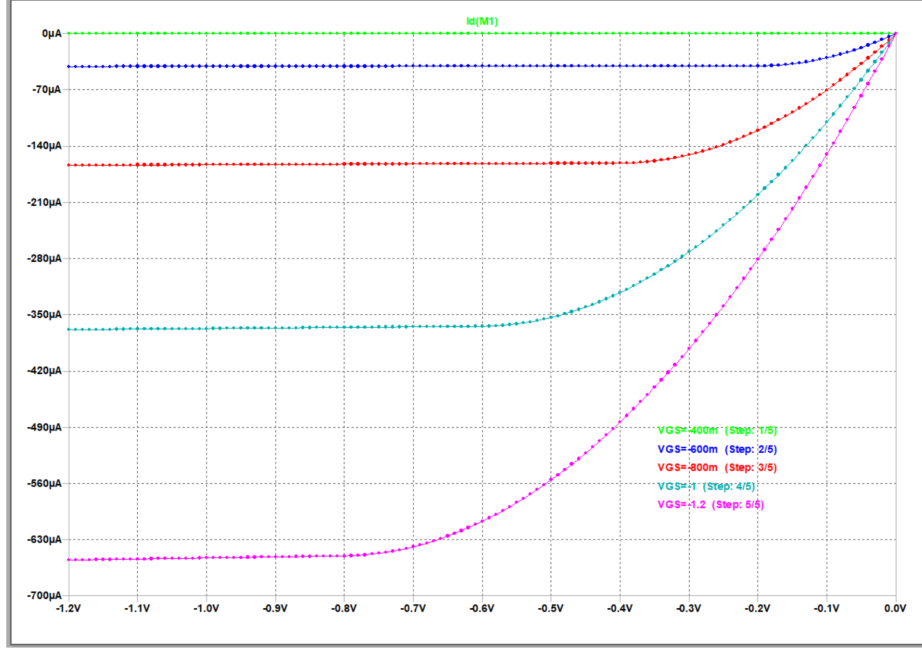


Figure 3: PMOS I-V Characteristics for various V_{GS} values.

From **Figure 3**, we observe five curves corresponding to $V_{GS} = -0.4V, -0.6V, -0.8V, -1.0V$, and $-1.2V$.

The curves mirror NMOS behaviour but appear in the third quadrant. Key observations are as follows.

In the linear region, the drain current increases steeply for small values of $|V_{DS}|$. In the saturation region, the current reaches a plateau when

$$|V_{DS}| \geq |V_{GS}| - |V_{th}|. \quad (5)$$

For verification with $V_{GS} = -1.2V$, the drain current magnitude is

$$I_D = \frac{\beta}{2}(V_{SG} - |V_{th}|)^2 \quad (6)$$

$$I_D = \frac{2000}{2}(1.2 - 0.4)^2 = 640 \mu A.$$

The magnitude matches that of the NMOS device because identical K_P and W/L ratios were used. In practical CMOS technologies, PMOS devices typically have a K_P value approximately half that of NMOS due to lower hole mobility. As a result, PMOS transistors are often designed to be approximately twice as wide to achieve matched drive strength.

2.3 Task 3, Gain Factor Consolidation and Comparison

Procedure and Theory

This task investigates how transistor geometry affects the drain current through the gain factor β . From Tasks 1 and 2, the saturation drain current is given by

$$I_D = \frac{\beta}{2}(V_{GS} - V_{th})^2(1 + \lambda V_{DS}). \quad (7)$$

The gain factor β is defined as

$$\beta = \mu_n C_{ox} \frac{W}{L} = K_P \frac{W}{L}. \quad (8)$$

Substituting this expression for β into the saturation current equation yields

$$I_D = \frac{K_P}{2} \cdot \frac{W}{L} \cdot (V_{GS} - V_{th})^2(1 + \lambda V_{DS}). \quad (9)$$

For fixed values of V_{GS} , V_{DS} , and L , this expression simplifies to

$$I_D = k \cdot W, \quad (10)$$

Question 1

From the *.param* lines:

$$V_{GS} = 1.2V \quad V_{DS} = 1.2V \quad L = 1\mu m$$

where k is a constant. This result predicts a linear relationship between the drain current I_D and the transistor width W , which is verified through simulation.

VTO (Threshold Voltage): Symbol: V_T or V_{th} . Units: Volts (V).

Zero-bias threshold voltage the minimum gate-source voltage required to form a conducting inversion channel. Below this voltage, the transistor is in cutoff. Determined by gate material work function, oxide thickness, substrate doping, and interface charges. In our simulation: VTO = 0.4V (NMOS), -0.4V (PMOS).

KP (Transconductance Parameter): Symbol: k' or μC_{ox} . Units: A/V².

Intrinsic transconductance parameter representing the product of carrier mobility (μ) and gate oxide capacitance per unit area (C_{ox}):

$$KP = \mu \cdot C_{ox} = \mu \cdot \frac{\epsilon_{ox}}{t_{ox}} \quad (11)$$

Determines current drive capability. The gain factor is $\beta = (W/L) \cdot KP$, so $I_D \propto KP$. NMOS typically has 2-3× higher KP than PMOS due to higher electron mobility. In our simulation: KP = 200μA/V² for both types.

LAMBDA (Channel-Length Modulation): Symbol: λ . Units: V⁻¹. Default: 0.0.

Channel-length modulation coefficient accounting for effective channel shortening as V_{DS} increases in saturation. Without LAMBDA, saturation curves would be perfectly flat (infinite output resistance). With LAMBDA:

$$I_D(sat) = \frac{\beta}{2}(V_{GS} - V_T)^2(1 + \lambda V_{DS}) \quad (12)$$

Output resistance: $r_o = 1/(\lambda I_D)$. Inversely proportional to channel length: $\lambda \propto 1/L$. Analogous to inverse Early voltage in BJTs. In our simulation: $\lambda = 0.02 \text{ V}^{-1}$, causing $\sim 2.4\%$ current increase across 1.2V V_{DS} range.

Code and Explanation

```

1 * Task 3: Beta Gain Factor Sweep
2 .param VGSVAL=1.2 VDSVAL=1.2
3 .param LVAL=1u
4
5 VGS G 0 DC {VGSVAL}
6 VDS D 0 DC {VDSVAL}
7
8 M1 D G 0 0 NLEVEL1 W={WVAL} L={LVAL}
9
10 .model NLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
11
12 .step param WVAL 2u 20u 2u          * Sweep W from 2um to 20um
13 .op                                  * DC operating point analysis
14
15 .end

```

Listing 3: PMOS Sweep Simulation Code

The `.step` command varies W while `.op` calculates the DC operating point for each value. Results appear in the SPICE log rather than as a plot.

Question 2B

The standard textbook equation for saturation current is:

$$I_D = \frac{\beta}{2}(V_{GS} - V_T)^2 \quad \text{where } \beta = \frac{W}{L} \cdot \mu_n C_{ox} \quad (13)$$

The complete SPICE Level 1 equation is:

$$I_D = \frac{KP}{2} \cdot \frac{W_{eff}}{L_{eff}} \cdot (V_{GS} - V_{TO})^2 \cdot (1 + LAMBDA \cdot V_{DS}) \quad (14)$$

To simplify the SPICE model to match textbook equations, the following assumptions are made:

1. LAMBDA = 0 (or very small): The term $(1 + \lambda V_{DS})$ is set to 1, neglecting channel-length modulation. This assumes perfectly flat I-V curves in saturation with infinite output resistance. In our simulation, $\lambda = 0.02$ causes only a $\sim 2.4\%$ variation, which is small but non-zero.

2. No body effect ($\gamma = 0$ or $V_{SB} = 0$): The body effect modifies threshold voltage as:

$$V_T = V_{T0} + \gamma(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}) \quad (15)$$

When bulk and source are both at ground ($V_{SB} = 0$), this term vanishes and $V_T = V_{T0}$, which is our case.

3. Long-channel approximation: Velocity saturation is neglected, assuming drift velocity remains proportional to electric field. Valid for $L \geq 1\mu\text{m}$. Short-channel effects (DIBL, threshold roll-off) are ignored.

4. No mobility degradation: Carrier mobility μ is assumed constant, independent of V_{GS} . In reality, high vertical electric fields reduce mobility, but Level 1 ignores this.

5. Ideal effective dimensions: $W_{eff} = W$ and $L_{eff} = L$ (no parasitic effects). When $LD = WD = XL = XW = DEL = 0$, drawn and effective dimensions are equal.

6. Subthreshold conduction ignored: For $V_{GS} < V_T$, Level 1 assumes $I_D = 0$. In reality, exponential subthreshold current exists: $I_D \propto \exp(V_{GS}/nV_t)$.

Verification of equivalence: If $KP = \mu_n C_{ox}$, then:

$$\beta_{textbook} = \frac{W}{L} \cdot \mu_n C_{ox} = \frac{W}{L} \cdot KP = \beta_{SPICE} \quad (16)$$

With $LAMBDA = 0$, $V_{SB} = 0$, and ideal geometry, the equations become identical.

$$I_D = \frac{K_P}{2} \frac{W}{L} (V_{GS} - V_{th})^2, \quad (17)$$

with no body-effect corrections included.

Results and Discussion

```

1 * --- Fixed Bias Parameters ---
2 .param VGSVAL=1.2
3 .param VDSVAL=1.2
4 .param LVAL= 1u
5
6 * --- Voltage Sources ---
7 VGS G 0 DC {VGSVAL}
8 VDS D 0 DC {VDSVAL}
9
10 * --- NMOS with L=2um (doubled) ---
11 M1 D G 0 0 NLEVEL1 W={WVAL} L={LVAL}
12
13 * --- Model ---
14 .model NLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
15
16 * --- Sweep W ---
17 .step param WVAL 2u 20u 2u
18
19 * --- Analysis ---
20 .op
21 .print OP I(M1)
22
23 .end

```

Listing 4: Beta Sweep Simulation Code

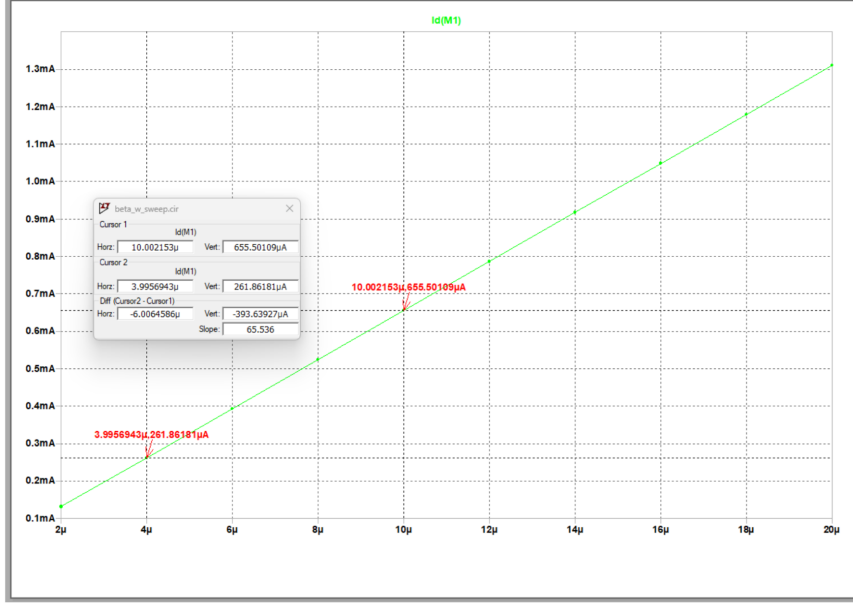


Figure 4: Beta Sweep

Figure 4, shows the linear relationship between drain current I_D and transistor width W . Each data point corresponds to a different width from $2\mu\text{m}$ to $20\mu\text{m}$.

The theoretical slope can be derived from the saturation drain current expression using the given parameters,

$$I_D = \frac{K_P}{2L}(V_{GS} - V_{th})^2(1 + \lambda V_{DS}) \cdot W. \quad (18)$$

Differentiating with respect to the transistor width W gives the slope,

$$\text{Slope} = \frac{dI_D}{dW} = \frac{K_P}{2L}(V_{GS} - V_{th})^2(1 + \lambda V_{DS}). \quad (19)$$

Substituting the values $K_P = 200 \mu\text{A}/\text{V}^2$, $L = 1 \mu\text{m}$, $V_{GS} = 1.2 \text{ V}$, $V_{th} = 0.4 \text{ V}$, $\lambda = 0.02 \text{ V}^{-1}$, and $V_{DS} = 1.2 \text{ V}$,

$$\text{Slope} = \frac{200 \mu}{2 \times 1 \mu}(1.2 - 0.4)^2(1 + 0.02 \times 1.2). \quad (20)$$

Evaluating,

$$\text{Slope} = 100 \times 0.64 \times 1.024 = 65.54 \mu\text{A}/\mu\text{m}. \quad (21)$$

From simulation, taking two points such as $W = 20 \mu\text{m}$ with $I_D \approx 1310 \mu\text{A}$ and $W = 2 \mu\text{m}$ with $I_D \approx 131 \mu\text{A}$,

$$\text{Slope} = \frac{1310 - 131}{20 - 2} = \frac{1179}{18} \approx 65.5 \mu\text{A}/\mu\text{m}. \quad (22)$$

This closely matches the theoretical prediction, confirming that $I_D \propto W$.

Doubling Channel Length

If the channel length is doubled to $L = 2\ \mu\text{m}$, then from the definition of the gain factor,

$$\beta = K_P \frac{W}{L}. \quad (23)$$

The new gain factor becomes

$$\beta_{\text{new}} = K_P \frac{W}{2L} = \frac{\beta_{\text{original}}}{2}. \quad (24)$$

As a result, the drain current I_D is halved for all values of W . Consequently, the slope of the I_D versus W characteristic becomes

$$\text{New Slope} = \frac{65.54}{2} = 32.77\ \mu\text{A}/\mu\text{m}. \quad (25)$$

This result illustrates why shorter channel lengths are preferred in transistor design to achieve higher drive current.

Results and Discussion

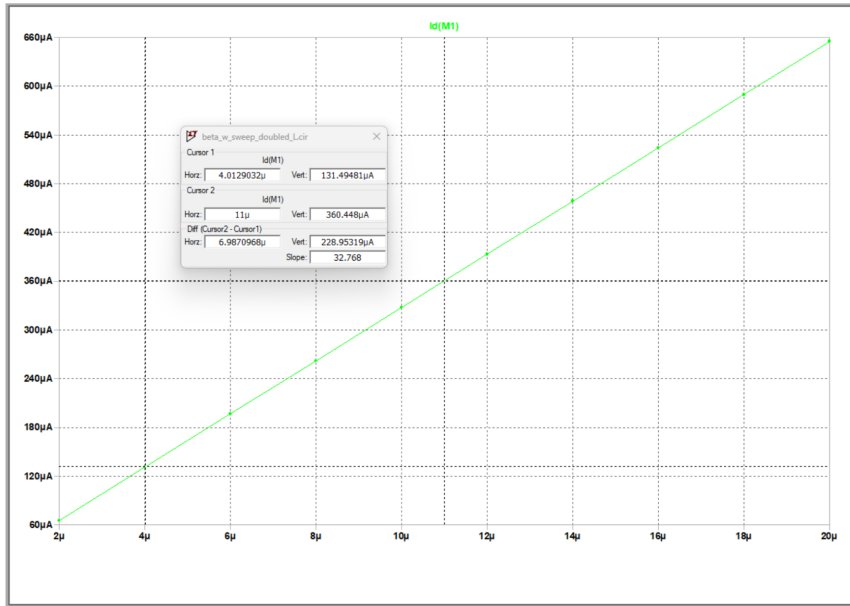


Figure 5: PMOS I-V Characteristics for various V_{GS} values.

Figure 5 shows the linear relationship between drain current I_D and transistor width W when the channel length is doubled to $L = 2\ \mu\text{m}$. Each data point corresponds to a different width from $2\ \mu\text{m}$ to $20\ \mu\text{m}$.

Reducing VGS

When $V_{GS} < V_{th}$, for example $V_{GS} = 0.3 \text{ V} < 0.4 \text{ V}$, the transistor operates in the cutoff region. In this condition, no inversion layer is formed and the drain current satisfies

$$I_D \approx 0, \quad (26)$$

independent of the transistor width W . This behaviour explains why the threshold voltage defines the effective on/off boundary in digital switching applications.

Why PMOS is wider than NMOS

In Task 2, equal values of K_P were used for NMOS and PMOS devices. In practice, however, the process transconductance parameters differ:

$$K_{P,n} = \mu_n C_{ox} \approx 200 \mu\text{A}/\text{V}^2, \quad (27)$$

$$K_{P,p} = \mu_p C_{ox} \approx 80\text{--}100 \mu\text{A}/\text{V}^2. \quad (28)$$

Since the hole mobility is lower than the electron mobility, with $\mu_p \approx 0.4\mu_n$, a PMOS transistor must be approximately 2–2.5 times wider than an NMOS transistor to achieve symmetric rise and fall times in a CMOS inverter.

Task 4, Iterative Solutions

SPICE solves circuits containing nonlinear devices, such as diodes and transistors, using iterative numerical methods. Unlike linear circuits, where Kirchhoff's laws lead to direct algebraic solutions, nonlinear device equations require iterative root-finding techniques.

Applying Kirchhoff's Voltage Law (KVL) to the series circuit gives

$$V_s - IR - V_d = 0. \quad (29)$$

The diode current is described by the Shockley equation,

$$I = I_s (e^{V_d/nV_t} - 1). \quad (30)$$

Substituting this expression for I into the KVL equation yields

$$f(V_d) = V_s - RI_s (e^{V_d/nV_t} - 1) - V_d = 0. \quad (31)$$

This is a nonlinear equation in V_d and cannot be solved analytically.

To find the root of $f(V_d) = 0$, SPICE employs the Newton–Raphson method, which iterates according to

$$V_d^{(k+1)} = V_d^{(k)} - \frac{f(V_d^{(k)})}{f'(V_d^{(k)})}. \quad (32)$$

The derivative of $f(V_d)$ is

$$f'(V_d) = -\frac{RI_s}{nV_t} e^{V_d/nV_t} - 1. \quad (33)$$

Convergence is achieved when the change in voltage satisfies $|\Delta V_d| < \varepsilon$, where ε is a chosen tolerance.

In earlier tasks, the Level 1 MOSFET model was used, which also involves nonlinear equations such as

$$I_D = \frac{\beta}{2}(V_{GS} - V_{th})^2. \quad (34)$$

When SPICE computes the DC operating point, it applies the NewtonRaphson method internally to solve these nonlinear device equations simultaneously with the Kirchhoff current and voltage constraints.

Parameter	Value
V_s	5 V
R	1 k Ω
I_s	10^{-12} A
n	1
V_t	26 mV (at $T = 300$ K)

Table 1: Circuit and diode parameters used in the analysis

Due to the length of the code, use the link to view the Python script here: (INSERT PYTHON CODE LINK GITHUB HERE).

Figure 6 shows the terminal output from running the NewtonRaphson iterative solver for the diode circuit. The iterations converge to a diode voltage of approximately 0.58 V, yielding a current of about 4.4 mA.

```

(base) sahasht@sahasht-Zenbook-S-13-UX5304VA-U
27_Project/ICDesign/Code/PythonSolutionCode/

Newton-Raphson Iteration Results:

Iteration      Vd (V)      Change (V)
1 6.743050e-01 2.569504e-02
2 6.489609e-01 2.534402e-02
3 6.245485e-01 2.441248e-02
4 6.024855e-01 2.206293e-02
5 5.856140e-01 1.687148e-02
6 5.770664e-01 8.547596e-03
7 5.753132e-01 1.753236e-03
8 5.752515e-01 6.172715e-05
9 5.752514e-01 7.324502e-08

Final Results:
Diode voltage (Vd): 0.575251 V
Diode current (I): 4.424749e-03 A

```

Figure 6: Newton Raphson Terminal Output

A cobweb diagram can be used to visualise the convergence of the NewtonRaphson iterations. **Figure 7** illustrates how successive approximations approach the root of the nonlinear equation.

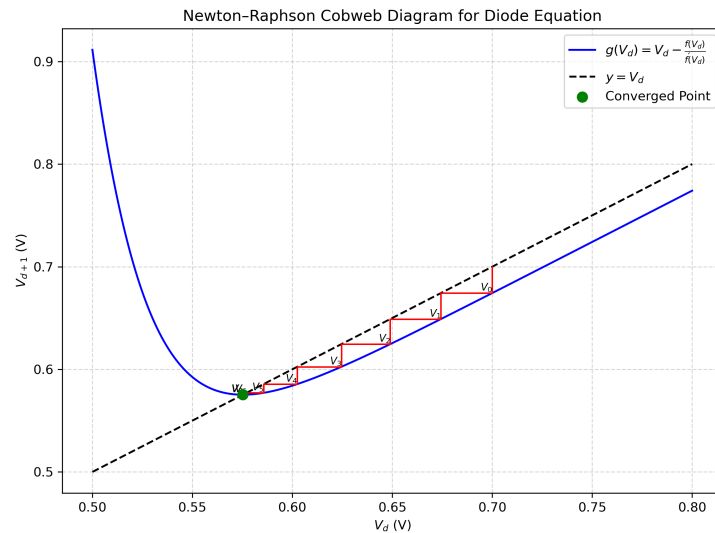


Figure 7: Cobweb Diagram Showing the Convergence of NewtonRaphson Iterations.

Results and Discussion

The Newton–Raphson method converges rapidly due to its quadratic convergence property, meaning that the number of correct digits approximately doubles with each iteration. From the numerical output, six-digit accuracy is achieved in approximately seven iterations. This computational efficiency is the reason SPICE employs the Newton–Raphson algorithm for DC operating-point analysis.

Verification of the solution can be performed using Kirchhoff's Voltage Law,

$$V_s - IR - V_d = 5 - (4.383 \times 10^{-3})(1000) - 0.6166 = 0, \quad (35)$$

which confirms that the computed values satisfy the circuit constraints.

When SPICE analyses the MOSFET circuits from Tasks 1–3, it formulates nonlinear equations from the device models, such as $I_D = f(V_{GS}, V_{DS})$, and combines them with the Kirchhoff voltage and current constraints.

These equations are linearised through construction of the Jacobian matrix, which is the multivariable generalisation of the derivative $f'(x)$.

SPICE then iterates using the NewtonRaphson method until all node voltages converge.

The `.OPTIONS` command in SPICE controls convergence parameters such as `RELTOL` (relative tolerance) and `ITL1` (DC iteration limit), which are directly analogous to the tolerance ε used in the Python-based implementation.

2.4 Lab 2

Lab 2: CMOS Inverter Design and Analysis

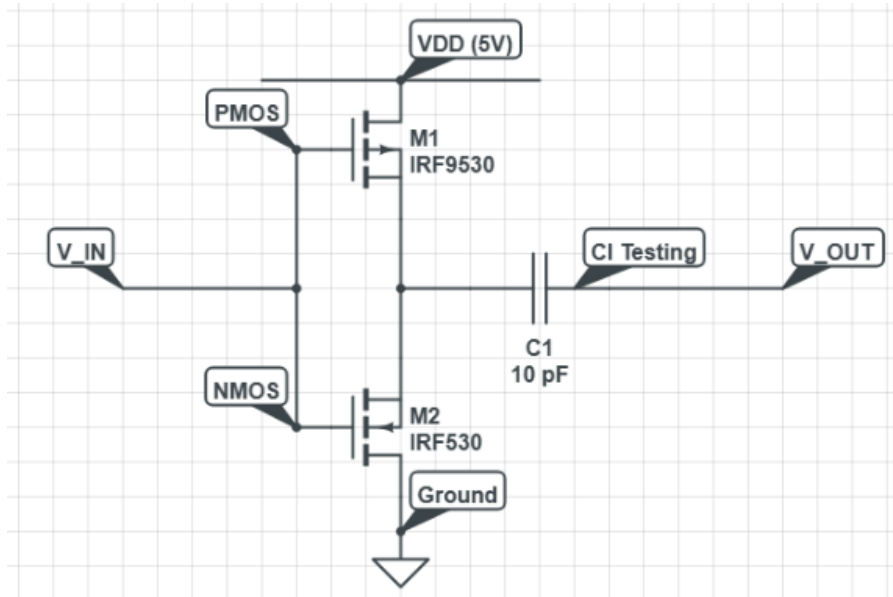


Figure 8: Cobweb Diagram Showing the Convergence of NewtonRaphson Iterations.

PMOS (pull-up network) $20\ \mu\text{m}$, $1\ \mu\text{m}$

NMOS (pull-down network) $10\ \mu\text{m}$, $1\ \mu\text{m}$

added here as it would not be clear in the image.

SPICE Template Completion

The SPICE netlist defines the circuit topology, device models, and analysis type. For DC analysis, we sweep V_{in} from 0V to 5V to observe the transfer characteristic.

The netlist code is as follows:

```

1 * Power supply: 5V DC connected between Vdd node and ground
2 VDD Vdd 0 DC 5
3
4 * Input voltage source: swept from 0 to 5V for DC analysis
5 Vin in 0 DC 0
6
7 * NMOS transistor M1 (pull-down network)
8 * Terminals: Drain=out, Gate=in, Source=0, Bulk=0
9 M1 out in 0 0 NMOSLEVEL1 W=10u L=1u
10
11 * PMOS transistor M2 (pull-up network)
12 * Terminals: Drain=out, Gate=in, Source=Vdd, Bulk=Vdd
13 M2 out in Vdd Vdd PMOSLEVEL1 W=20u L=1u
14
15 * DC sweep: Vin from 0V to 5V in 10mV steps
16 .dc Vin 0 5 0.01
17
18 * Plot input and output voltages
19 .plot DC V(in) V(out)
20
21 * NMOS Level 1 model: Vth=0.4V, KP=200uA/V^2, lambda=0.02
22 .model NMOSLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
23
24 * PMOS Level 1 model: Vth=-0.4V, KP=80uA/V^2, lambda=0.02
25 .model PMOSLEVEL1 PMOS LEVEL=1 VTO=-0.4 KP=80u LAMBDA=0.02
26
27 .end

```

Listing 5: SPICE template completion

The netlist defines complementary NMOS and PMOS transistors that share a common gate (input) and a common drain (output). The PMOS device is sized with a width of $W = 20\ \mu\text{m}$, which is twice that of the NMOS transistor, to compensate for the lower hole mobility. This sizing choice ensures approximately symmetric rise and fall times during switching.

1C: DC Transfer Curve

The DC transfer curve shows the static relationship between input and output. Five operating regions exist based on which transistor is in cutoff, linear, or saturation.

At point A, with $V_{\text{in}} = 0\text{ V}$, the NMOS transistor has $V_{GS} = 0 < V_{th}$ and is therefore turned off. The PMOS transistor has $V_{SG} = 5\text{ V} > |V_{th}|$, conducts strongly, and pulls the output voltage such that $V_{\text{out}} \approx V_{DD}$.

At point C, with $V_{\text{in}} = 2.5\text{ V}$, both transistors operate in saturation. This corresponds to the high-gain transition region, where small changes in the input voltage produce large swings in the output voltage.

At point E, with $V_{\text{in}} = 5\text{ V}$, the PMOS transistor has $V_{SG} = 0 < |V_{th}|$ and is therefore turned off. The NMOS transistor conducts, pulling the output voltage such that $V_{\text{out}} \approx 0\text{ V}$.

The transfer curve can be seen clearly below:

Point	V_{in} (V)	V_{out} (V)	NMOS Mode	PMOS Mode
A	0	~ 5	Cutoff	Linear
B	1	~ 5	Saturation	Linear
C	2.5	~ 2.5	Saturation	Saturation
D	4	~ 0	Linear	Saturation
E	5	~ 0	Linear	Cutoff

Table 2: Operating regions of NMOS and PMOS transistors at key points on the CMOS inverter voltage transfer characteristic

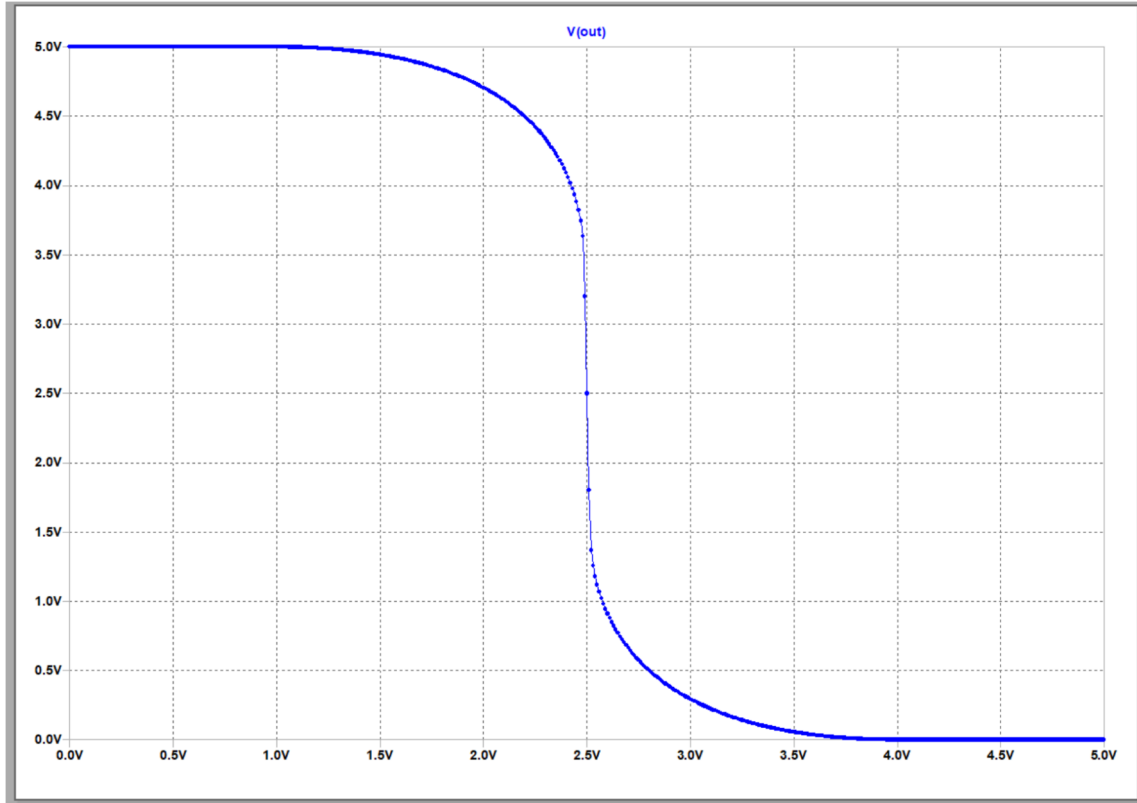


Figure 9: CMOS Transfer Curve

The following figures show the cursor positions for points A through E on the voltage transfer curve.

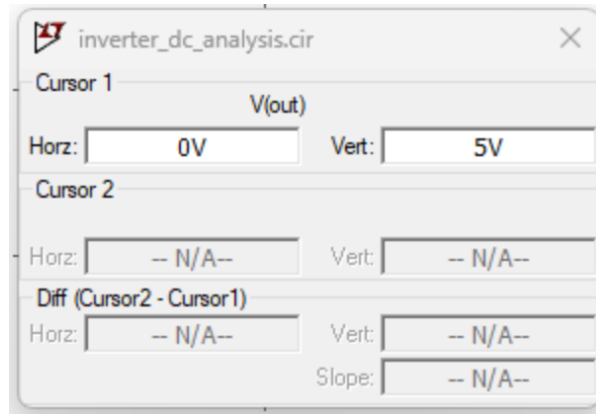


Figure 10: CMOS Point A

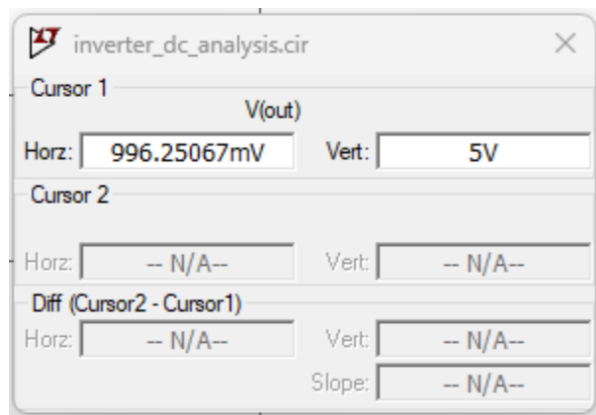


Figure 11: CMOS Point B

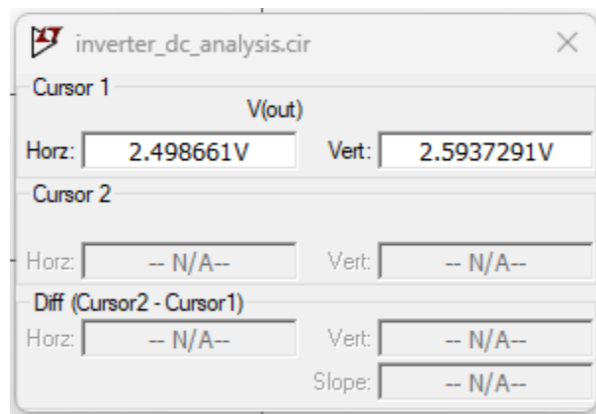


Figure 12: CMOS Point C

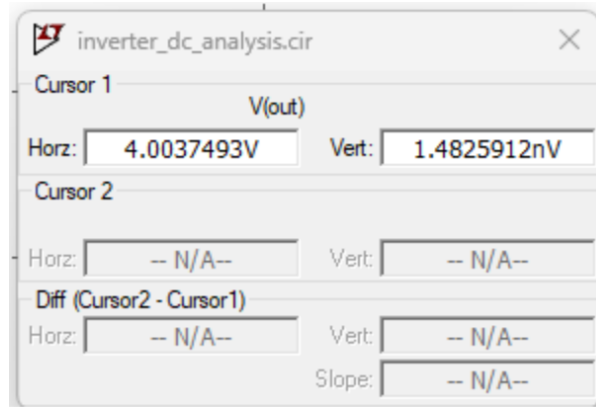


Figure 13: CMOS Point D

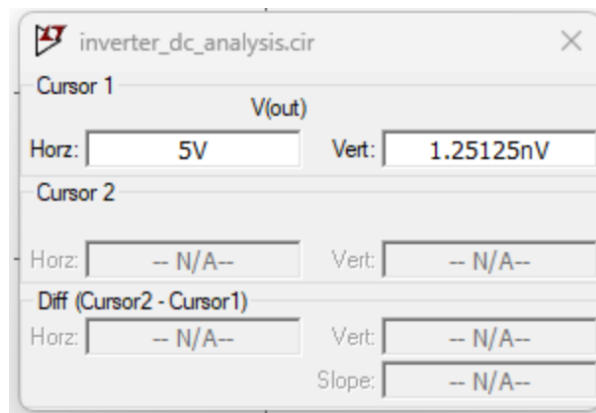


Figure 14: CMOS Point E

1D NMOS Pull Up Exploration

By altering the CMOS structure to only use NMOS transistors for both pull-up and pull-down networks, we can observe the impact on the voltage transfer characteristic. The modified netlist is as follows:

```

1 * NMOS-only Inverter (Pseudo-NMOS)
2 VDD Vdd 0 DC 5
3 Vin in 0 DC 0
4
5 * Pull-down NMOS
6 M1 out in 0 0 NMOSLEVEL1 W=10u L=1u
7
8 * Pull-up NMOS (gate tied to VDD - always trying to conduct)
9 M2 out Vdd Vdd 0 NMOSLEVEL1 W=20u L=1u
10
11 .dc Vin 0 5 0.01
12 .plot DC V(out)
13 .model NMOSLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
14 .end

```

Listing 6: NMOS Pull Down

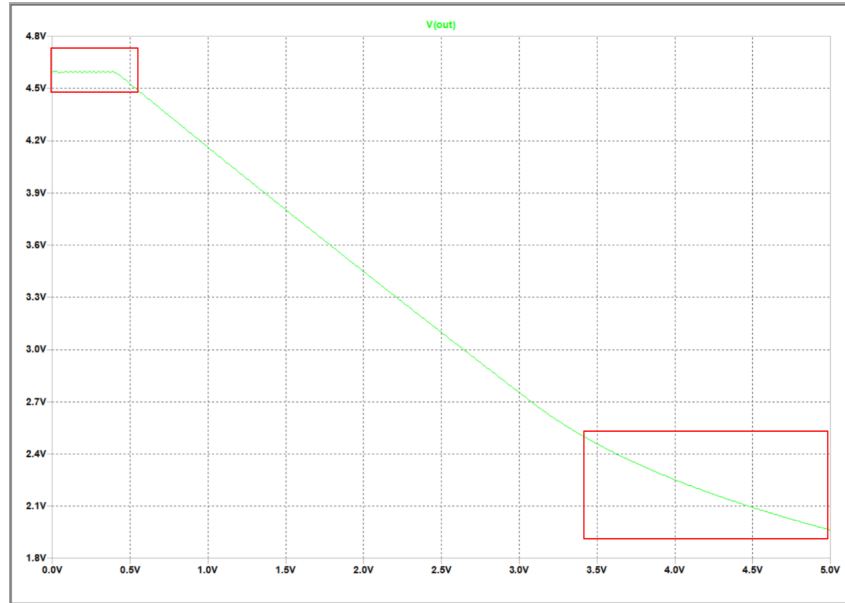


Figure 15: NMOS Pull Down Transfer Curve

Q1: What is the highest output voltage? Why not 5V?

The maximum $V_{out} \approx 4.6V$ ($= V_{DD} - V_{th} = 5 - 0.4V$). The NMOS pull-up cannot pass a voltage higher than $V_{GS} - V_{th}$. When V_{out} approaches $V_{DD} - V_{th}$, the pull-up NMOS enters cutoff because $V_{GS} = V_{DD} - V_{out}$ drops below V_{th} .

Q2: What happens to the NMOS pull-up when $V_{in} = 0$?

When $V_{in} = 0V$, the pull-down M1 is OFF. The pull-up M2 tries to charge the output, but it's operating as a source follower with $V_{GS} = V_{DD} - V_{out}$. As V_{out} rises, V_{GS} decreases until $V_{GS} = V_{th}$, at which point M2 enters cutoff and charging stops.

Q3: What happens if this drives another CMOS gate?

The degraded logic '1' (4.6V instead of 5V) may not fully turn OFF the PMOS in the next stage. This causes:

- Static power dissipation (both transistors partially ON)
- Reduced noise margin
- Slower switching in subsequent stages

Q4: Why does PMOS avoid this issue?

PMOS passes strong '1's because its source connects to V_{DD} . When pulling up, $V_{SG} = V_{DD} - V_{out}$ remains high until $V_{out} = V_{DD}$. PMOS can fully charge the output to V_{DD} without the threshold voltage drop limitation.

From the figure above, the appropriate regions have been labelled.

Task 2: Inverter Transient Analysis

Transient analysis characterises the dynamic switching behaviour of the circuit. The key timing parameters are defined as follows.

$$t_{\text{rise}} = t_{90\%} - t_{10\%}, \quad (36)$$

$$t_{\text{fall}} = t_{10\%} - t_{90\%}, \quad (37)$$

$$t_{PLH} = t_{\text{out},50\%} - t_{\text{in},50\%} \quad (\text{output falling}), \quad (38)$$

$$t_{PLH} = t_{\text{out},50\%} - t_{\text{in},50\%} \quad (\text{output rising}). \quad (39)$$

The design specifications are:

$$t_{\text{rise}} \leq 15 \text{ ns},$$

$$t_{\text{fall}} \leq 10 \text{ ns},$$

$$|t_{PLH} - t_{PHL}| \leq 2 \text{ ns}.$$

Code and Explanation

```

1 * CMOS Inverter Transient - Original Design
2 VDD Vdd 0 DC 5
3 Vin in 0 PULSE(0 5 0ns 5ns 5ns 50ns 100ns) * 100ns period pulse
4
5 M1 out in 0 0 NMOSLEVEL1 W=10u L=1u
6 M2 out in Vdd Vdd PMOSLEVEL1 W=20u L=1u
7 CL out 0 10p * 10pF load
8
9 .tran 0.1ns 200ns
10
11 .model NMOSLEVEL1 NMOS LEVEL=1 VTO=0.4 KP=200u LAMBDA=0.02
12 .model PMOSLEVEL1 PMOS LEVEL=1 VTO=-0.4 KP=80u LAMBDA=0.02
13
14 * Automated measurements
15 .meas tran t_rise TRIG V(out) VAL=0.5 RISE=1 TARG V(out) VAL=4.5 RISE=1
16 .meas tran t_fall TRIG V(out) VAL=4.5 FALL=1 TARG V(out) VAL=0.5 FALL=1
17 .meas tran tPHL TRIG V(in) VAL=2.5 RISE=1 TARG V(out) VAL=2.5 FALL=1
18 .meas tran tPLH TRIG V(in) VAL=2.5 FALL=1 TARG V(out) VAL=2.5 RISE=1
19 .meas tran t_skew PARAM='abs(tPHL-tPLH)'
20 .end

```

Listing 7: NMOS Pull Down

The PULSE source generates a $0 - 5V$ square wave with $5ns$ rise/fall times and $100ns$ period. The `.meas` commands automatically extract timing parameters.

```

LTspice 24.1.6 for Windows
Circuit: C:\Users\sahas\OneDrive\Documents\LTspice\NewLTspice\inverter_transient_with_meas.cir
Start Time: Sun Dec 28 13:36:03 2025
solver = Normal
Maximum thread count: 4
tnom = 27
temp = 27
method = trap
Instance "M1": Length shorter than recommended for a level 1 MOSFET.
Instance "M1": Width narrower than recommended for a level 1 MOSFET.
Instance "M2": Length shorter than recommended for a level 1 MOSFET.
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.440 seconds.

Files loaded:
C:\Users\sahas\OneDrive\Documents\LTspice\NewLTspice\inverter_transient_with_meas.cir

t_rise=4.31961111041e-09 FROM 5.81766831026e-08 TO 6.24962942131e-08
t_fall=3.72037502918e-09 FROM 2.94452195604e-09 TO 6.66489698522e-09
tphl=2.16419158955e-09 FROM 2.49999995702e-09 TO 4.66419154657e-09
tplh=2.5167216919e-09 FROM 5.74999999869e-08 TO 6.00167216788e-08
t_prop: '(tPHL+tPLH)/2'=2.34045664073e-09
t_skew: 'abs(tPHL-tPLH) '=3.5253010234e-10

```

Figure 16: Original Transient Analysis Log Output

FPGA DESIGN LABS (LABS 4, 5 & 6)

ASYNCHRONOUS CIRCUIT MODELLING WITH WORKCRAFT (LABS 7 & 8)

REFERENCES

APPENDIX
